

Hyunseok Hong

KAIST Computer Science undergraduate

contact@hyunseok.dev · github.com/ding-modding · hyunseok.dev · South Korea

EXPERIENCE

Research Intern — KIXLAB, KAIST Interaction Lab

Dec 2025 - Mar 2026

- Research project for Samsung Electronics — designed a classification taxonomy and experiment protocol for its generative-AI features.
- Joined VideoVerse — research on automatic cognitive-load analysis and adaptive editing of educational video for learners with intellectual disabilities.
- Co-designed and built, with my mentor, a framework for per-segment cognitive-complexity scoring via a Gemini VLM, plus the multi-stage pipeline that derives editing operations matched to each learner's ability gap.
- Owned the VLM + SAM object tracking-and-removal pipeline.

Team Lead — Teamplo · Mad on

Mar 2026 - Present

- A group-project tool where meeting notes, tasks, and calendar interlock. Pivoted from a personal to-do app to a team-focused tool after PO user interviews.
- Offers an AI feature that breaks existing tasks down into detailed subtasks.

Next.js · PostgreSQL · Yjs · CodeMirror 6

Team Lead — Timeschool · Mad on

Aug 2024 - Mar 2026

- Built and operated Timeschool, a night-study management web platform; signed a service contract with Incheon Jinsan Science High School, running 3 years.
- Led the Node.js backend and contributed the React/Vite frontend; iterated each semester on user feedback.

RESEARCH

Reddit vs Moltbook — KAIST

hyunseok.dev/research

- A distant-reading comparison of a human platform (Reddit) vs an AI-agent platform (Moltbook), measuring how human and AI data diverge across five dimensions: text, network, time, author activity, and the driving model. Presentation deck + results writeup published.
- Collected and analyzed ~150,000 posts and comments across the two platforms.

Distant reading · MTLD · VADER · Reply-tree analysis · KS statistic

Asynchronous Multi-Agent RL for Heterogeneous CPU Scheduling — KAIST, team of 3

Mar - Jun 2026

- Modeled heterogeneous (P-core / E-core) CPU scheduling as an asynchronous Multi-Agent RL problem — each core an agent under an Agent-Centric Actor-Critic (ACAC) policy on a MAPPO baseline.
- Designed the experimental environment and reward function, ran experiments, and analyzed results to iterate on the model.

SGLP — Sigmoidal Gradient Length Prompting — KAIST

Mar - Jun 2025

- Proposed and developed SGLP, a novel method unifying prior work's complex two-stage prompt learning (DPL) into a single end-to-end pipeline.
- Designed a sigmoid-based gating mechanism with bilevel optimization to autonomously discover optimal layer-wise prompt lengths.
- Achieved meaningful accuracy (91.83% on EuroSAT) while reducing trainable parameters $7\times$ ($0.258\text{M} \rightarrow 37\text{K}$).

A Study on Electric-Scooter Stability by Rider Posture Using Deep Learning — Incheon Jinsan Science High School

Mar - Jul 2023

- Designed and implemented a custom reinforcement-learning environment and reward system in Unity to simulate e-scooter dynamics.
- Applied deep reinforcement learning via the Unity ML-Agents framework to train the model.
- Proposed the optimal rider posture to improve stability and safety for e-scooters on inclines.
- Top Prize, Incheon Science Exhibition.

PROJECTS

MatchA — Core Developer

matchall.co.kr

– Group scheduling: a workspace that organizes groups and calendars like folders, with time-voting and When2Meet-style coordination.

Next.js · Vercel

EDUCATION

KAIST — School of Computing, B.S. in Computer Science

Feb 2024 - Present

– GPA 3.96 / 4.3 (82 credits)

– **Selected coursework** — CS.30701 Introduction to Deep Learning (A+) · CS.30706 Machine Learning (A+) · CS.30200 Programming Language (A+) · CS.30000 Introduction to Algorithms (A+) · CS.30101 Computer Organization (A+) · CS.20004 Discrete Mathematics (A0) · CS.20006 Data Structure (A0) · MAS.10009 Introduction to Linear Algebra (A+) · MAS.10001 Calculus I (A+) · MAS.10002 Calculus II (A+)

Incheon Jinsan Science High School — early graduation

Mar 2022 - Feb 2024

SKILLS

Languages	C, C++, C#, Python, Java, TypeScript, JavaScript
Frameworks & Tools	React, Next.js, Node.js, Tailwind CSS, PyTorch
Workflow	AI-assisted development (Claude Code, Codex)

HONORS & AWARDS

2025	NYPC Code Battle — top 4.87%
2024	Dean's List (1st & 2nd semester)
2024	KAIST President's Award
2023	Korean Olympiad in Informatics (KOI) — Silver, 12th nationally (high-school division)
2023	ICT Award Korea — Grand Prize · Incheon Science Exhibition — Top Prize
2022	CPS (Creative Problem Solving) Festival — Grand Prize

SCHOLARSHIP

2024	KAIST Presidential Fellowship (KPF) — a highly selective fellowship for top undergraduates: overseas summer programs at leading universities, undergraduate research (URP) priority, and leadership training & faculty mentoring
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OPEN SOURCE

gbrain (github.com/garrytan/gbrain, 17.8k★) — diagnosed why the OAuth public-client flow didn't follow the PKCE + RFC 7591 standards, causing auth failures for some clients; fix PR merged in v0.34.1.0.